

Solved selected problems of Calculus on Manifolds by Michael Spivak

Franco Zacco

2 - Differentiation

Basic Definitions

Proof. 2-2. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be independent of the second variable i.e. for each $x \in \mathbb{R}$ we have that $f(x, y_1) = f(x, y_2)$ for all $y_1, y_2 \in \mathbb{R}$.

Then we can take some $y_0 \in \mathbb{R}$ and define a function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(x) = f(x, y_0)$, then, for any $y \in \mathbb{R}$ we get that $f(x, y) = f(x, y_0) = g(x)$ because f is independent of the second variable.

On the other hand, suppose we have a function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(x) = f(x, y)$ then since this is true for any $y \in \mathbb{R}$ then we can write that

$$f(x, y_1) = g(x) = f(x, y_2)$$

for any $y_1, y_2 \in \mathbb{R}$, which implies that f is independent of the second variable.

The Jacobian of f at (a, b) i.e. $f'(a, b)$ is given by

$$f'(a, b) = (D_1f(a, b), D_2f(a, b))$$

Since $g(x) = f(x, y)$ then $D_1f(a, b) = D_1g(a) = g'(a)$ therefore

$$f'(a, b) = (g'(a), 0)$$

□

Proof. 2-3. A function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ is independent of the first variable if for each $y \in \mathbb{R}$ we have that $f(x_1, y) = f(x_2, y)$ for all $x_1, x_2 \in \mathbb{R}$.

The Jacobian $f'(a, b)$ in this case is

$$f'(a, b) = (D_1f(a, b), D_2f(a, b)) = (0, D_2f(a, b))$$

In the same way as we did in problem 2-2, we can show that f is independent of the first variable if and only if there is a function $g : \mathbb{R} \rightarrow \mathbb{R}$ such that $g(y) = f(x, y)$.

So we get that $D_2f(a, b) = D_2g(b) = g'(b)$ and therefore

$$f'(a, b) = (0, g'(b))$$

Finally, the functions that are independent of the first and the second variable are the functions of the form $f(x, y) = c$ where c is a constant. $\square$

Proof. 2-4. Let us define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x) = \begin{cases} |x| \cdot g\left(\frac{x}{|x|}\right) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Where g is a continuous real-valued function on the unit circle $\{x \in \mathbb{R}^2 : |x| = 1\}$ such that $g(0, 1) = g(1, 0) = 0$ and $g(-x) = -g(x)$.

(a) If $x \in \mathbb{R}^2$ and $h : \mathbb{R} \rightarrow \mathbb{R}$ defined by $h(t) = f(tx)$, we want to show that h is differentiable.

Suppose $x \neq 0$, also, let $a \in \mathbb{R}$ and let $\lambda(k) = k|x|g\left(\frac{x}{|x|}\right)$ then

$$\begin{aligned} \lim_{k \rightarrow 0} \frac{|h(a+k) - h(a) - \lambda(k)|}{|k|} &= \lim_{k \rightarrow 0} \frac{|f((a+k)x) - f(ax) - \lambda(k)|}{|k|} = \\ &= \lim_{k \rightarrow 0} \frac{1}{|k|} \left| |(a+k)x|g\left(\frac{(a+k)x}{|(a+k)x|}\right) - |ax|g\left(\frac{ax}{|ax|}\right) - k|x|g\left(\frac{x}{|x|}\right) | \right| \end{aligned}$$

Suppose $a+k > 0$, and suppose $a > 0$ then

$$\begin{aligned} &\frac{1}{|k|} \left| |(a+k)x|g\left(\frac{(a+k)x}{|(a+k)x|}\right) - |ax|g\left(\frac{ax}{|ax|}\right) - k|x|g\left(\frac{x}{|x|}\right) | \right| = \\ &= \frac{1}{|k|} \left| (a+k)|x|g\left(\frac{x}{|x|}\right) - a|x|g\left(\frac{x}{|x|}\right) - k|x|g\left(\frac{x}{|x|}\right) | \right| \\ &= 0 \end{aligned}$$

If $a < 0$ then

$$|ax|g\left(\frac{ax}{|ax|}\right) = -a|x|g\left(-\frac{ax}{a|x|}\right) = a|x|g\left(\frac{x}{|x|}\right)$$

Where we used that $g(-x) = -g(x)$ and hence

$$\begin{aligned} &\frac{1}{|k|} \left| |(a+k)x|g\left(\frac{(a+k)x}{|(a+k)x|}\right) - |ax|g\left(\frac{ax}{|ax|}\right) - k|x|g\left(\frac{x}{|x|}\right) | \right| = \\ &= \frac{1}{|k|} \left| (a+k)|x|g\left(\frac{x}{|x|}\right) - a|x|g\left(\frac{x}{|x|}\right) - k|x|g\left(\frac{x}{|x|}\right) | \right| \\ &= 0 \end{aligned}$$

Now suppose $a+k < 0$ then

$$\begin{aligned} &\frac{1}{|k|} \left| |(a+k)x|g\left(\frac{(a+k)x}{|(a+k)x|}\right) - |ax|g\left(\frac{ax}{|ax|}\right) - k|x|g\left(\frac{x}{|x|}\right) | \right| = \\ &= \left| -(a+k)|x|g\left(-\frac{(a+k)x}{(a+k)|x|}\right) - a|x|g\left(\frac{x}{|x|}\right) - k|x|g\left(\frac{x}{|x|}\right) \right| \\ &= \left| (a+k)|x|g\left(\frac{x}{|x|}\right) - a|x|g\left(\frac{x}{|x|}\right) - k|x|g\left(\frac{x}{|x|}\right) \right| \\ &= 0 \end{aligned}$$

Finally if $a + k = 0$

$$\begin{aligned} & \frac{1}{|k|} \left| |(a+k)x|g\left(\frac{(a+k)x}{|(a+k)x|}\right) - |ax|g\left(\frac{ax}{|ax|}\right) - k|x|g\left(\frac{x}{|x|}\right) \right| = \\ &= \frac{1}{|k|} \left| k|x|g\left(\frac{ax}{|ax|}\right) - k|x|g\left(\frac{x}{|x|}\right) \right| \\ &= 0 \end{aligned}$$

In the case $x = 0$ we see that $\lambda(k) = 0$ and hence

$$\begin{aligned} & \frac{1}{|k|} \left| |(a+k)x|g\left(\frac{(a+k)x}{|(a+k)x|}\right) - |ax|g\left(\frac{ax}{|ax|}\right) - k|x|g\left(\frac{x}{|x|}\right) \right| = \\ &= \frac{1}{|k|} \left| 0 - 0 - 0 \right| \\ &= 0 \end{aligned}$$

Therefore h is differentiable.

(b) Let us compute $Df(0,0)$ as follows

$$\begin{aligned} & \lim_{(h,k) \rightarrow (0,0)} \frac{|f(0+(h,k)) - f(0) - Df(0,0)(h,k)|}{|(h,k)|} \\ &= \lim_{(h,k) \rightarrow (0,0)} \frac{1}{|(h,k)|} \left| |(h,k)|g\left(\frac{(h,k)}{|(h,k)|}\right) - Df(0,0)(h,k) \right| \end{aligned}$$

Let $h = 0$ then

$$\begin{aligned} & \frac{1}{|(0,k)|} \left| |(0,k)|g\left(\frac{(0,k)}{|(0,k)|}\right) - Df(0,0)(0,k) \right| \\ &= \frac{1}{|k|} \left| |k|g\left(0, \frac{k}{|k|}\right) - D_2f(0,0)k \right| \\ &\leq \frac{1}{|k|} \left| |k|g\left(0, \frac{k}{|k|}\right) \right| + \frac{1}{|k|} \left| D_2f(0,0)k \right| \\ &= \left| g\left(0, \frac{k}{|k|}\right) \right| + \left| D_2f(0,0) \right| \end{aligned}$$

We see that $g(0, k/|k|) = g(0, 1) = 0$ if $k > 0$, if $k < 0$ then $g(0, k/|k|) = g(0, -1) = g(-(0, 1)) = -g(0, 1) = 0$, then we get that

$$0 \leq \frac{1}{|(0,k)|} \left| |(0,k)|g\left(\frac{(0,k)}{|(0,k)|}\right) - Df(0,0)(0,k) \right| \leq \left| D_2f(0,0) \right|$$

In the same way, if we let $k = 0$ instead

$$\begin{aligned}
& \frac{1}{|(h, 0)|} \left| |(h, 0)| g\left(\frac{(h, 0)}{|(h, 0)|}\right) - Df(0, 0)(h, 0) \right| \\
&= \frac{1}{|h|} \left| |h| g\left(\frac{h}{|h|}, 0\right) - D_1 f(0, 0)h \right| \\
&\leq \frac{1}{|h|} \left| |h| g\left(\frac{h}{|h|}, 0\right) \right| + \frac{1}{|h|} \left| D_1 f(0, 0)h \right| \\
&= \left| g\left(\frac{h}{|h|}, 0\right) \right| + \left| D_1 f(0, 0) \right|
\end{aligned}$$

Then since $g(0, 1) = 0$ we get that

$$0 \leq \frac{1}{|(h, 0)|} \left| |(h, 0)| g\left(\frac{(h, 0)}{|(h, 0)|}\right) - Df(0, 0)(h, 0) \right| \leq \left| D_1 f(0, 0) \right|$$

Therefore must be that $Df(0, 0) = 0$ for the limit

$$\lim_{(h, k) \rightarrow (0, 0)} \frac{|f(0 + (h, k)) - f(0) - Df(0, 0)(h, k)|}{|(h, k)|}$$

to converge to 0.

Then for any path taken by h and k the limit becomes

$$\begin{aligned}
& \lim_{(h, k) \rightarrow (0, 0)} \frac{|f(0 + (h, k)) - f(0) - Df(0, 0)(h, k)|}{|(h, k)|} \\
&= \lim_{(h, k) \rightarrow (0, 0)} \frac{1}{|(h, k)|} \left| |(h, k)| g\left(\frac{(h, k)}{|(h, k)|}\right) - 0 \right| \\
&= \lim_{(h, k) \rightarrow (0, 0)} \left| g\left(\frac{(h, k)}{|(h, k)|}\right) \right|
\end{aligned}$$

But g is only defined at the unit circle so this limit can only be 0 if $g = 0$ and hence f is not continuous at $(0, 0)$.

□

Proof. 2-5. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} \frac{x|y|}{\sqrt{x^2+y^2}} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases}$$

We can write that

$$\frac{x|y|}{\sqrt{x^2+y^2}} = \sqrt{x^2+y^2} \cdot \frac{x|y|}{x^2+y^2} = |(x, y)| \cdot \frac{x|y|}{|(x, y)|^2}$$

Then defining g as $g(x, y) = x|y|$ we see that

$$g\left(\frac{(x, y)}{|(x, y)|}\right) = \frac{x}{|(x, y)|} \left| \frac{y}{|(x, y)|} \right| = \frac{x|y|}{|(x, y)|^2}$$

Also, we see that $g(0, 1) = g(1, 0) = 0$ and that

$$g(-x, -y) = -x|-y| = -xy = -x|y| = -g(x, y)$$

Hence we can write $f(x, y)$ as

$$f(x, y) = \begin{cases} |(x, y)| \cdot g\left(\frac{(x, y)}{|(x, y)|}\right) & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases}$$

Where g is a continuous real-valued function with the same properties described in Problem 2-4.

Therefore we can conclude that f is not differentiable at $(0, 0)$. $\square$

Proof. 2-6. Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by $f(x, y) = \sqrt{|xy|}$, we may also define f as

$$f(x, y) = \begin{cases} \frac{\sqrt{x^2+y^2}\sqrt{|xy|}}{\sqrt{x^2+y^2}} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases}$$

We see that

$$\frac{\sqrt{x^2+y^2}\sqrt{|xy|}}{\sqrt{x^2+y^2}} = |(x, y)| \frac{\sqrt{|xy|}}{|(x, y)|}$$

We can define a function g as $g(x, y) = \sqrt{|xy|}$ so

$$g\left(\frac{(x, y)}{|(x, y)|}\right) = \sqrt{\left|\frac{x}{|(x, y)|} \frac{y}{|(x, y)|}\right|} = \sqrt{\frac{|xy|}{|(x, y)|^2}} = \frac{\sqrt{|xy|}}{|(x, y)|}$$

Also, we see that $g(0, 1) = g(1, 0) = 0$ and that

$$g(-x, -y) = \pm\sqrt{|(-x)(-y)|} = \pm\sqrt{|xy|} = \pm g(x, y)$$

Hence we can write $f(x, y)$ as

$$f(x, y) = \begin{cases} |(x, y)| \cdot g\left(\frac{(x, y)}{|(x, y)|}\right) & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases}$$

Where g is a continuous real-valued function with the same properties described in Problem 2-4.

Therefore we can conclude that f is not differentiable at $(0, 0)$. $\square$

Proof. 2-7. Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a function such that $|f(x)| \leq |x|^2$ we want to show that f is differentiable at 0.

Let us consider the following limit

$$\lim_{h \rightarrow 0} \frac{|f(0+h) - f(0) - Df(0)h|}{|h|}$$

We see that

$$\begin{aligned} 0 \leq \frac{|f(0+h) - f(0) - Df(0)h|}{|h|} &\leq \frac{|f(h)| + |f(0)| + |Df(0)h|}{|h|} \\ &\leq \frac{|h|^2 + 0 + |Df(0)||h|}{|h|} \\ &\leq |h| + |Df(0)| \end{aligned}$$

We see that $|h| + |Df(0)| \rightarrow 0$ as $h \rightarrow 0$ if $Df(0) = 0$.

Therefore f is differentiable at 0. □

Basic Theorems

Proof. 2-12

Let $f : \mathbb{R}^n \times \mathbb{R}^m \rightarrow \mathbb{R}^p$ be a bilinear function.

(a) We want to prove that $\lim_{(h,k) \rightarrow 0} |f(h,k)|/|(h,k)| = 0$, we note that

$$f(h,k) = \sum_{i=1}^n \sum_{j=1}^m h^i k^j f(e_i, e_j)$$

Where h^i and k^j are the components of h and k respectively. Then we see that

$$\frac{|f(h,k)|}{|(h,k)|} \leq \frac{1}{|(h,k)|} \sum_{i=1}^n \sum_{j=1}^m |h^i k^j f(e_i, e_j)| \leq \frac{1}{|(h,k)|} \sum_{i=1}^n \sum_{j=1}^m |h^i| |k^j| |f(e_i, e_j)|$$

Let us take $M = \max_{i,j} |f(e_i, e_j)|$ then we get that

$$\frac{|f(h,k)|}{|(h,k)|} \leq \frac{1}{|(h,k)|} \sum_{i=1}^n \sum_{j=1}^m |h^i| |k^j| |f(e_i, e_j)| \leq \frac{M}{|(h,k)|} \sum_{i=1}^n \sum_{j=1}^m |h^i| |k^j|$$

Now, given that $|h^i| |k^j| \leq |h| |k|$ we have that

$$\begin{aligned} M \frac{1}{|(h,k)|} \sum_{i=1}^n \sum_{j=1}^m |h^i| |k^j| &\leq M \frac{nm|h||k|}{\sqrt{|h|^2 + |k|^2}} \\ &\leq M \frac{nm|h||k|}{\max\{|h|, |k|\}} \\ &\leq Mnm \min\{|h|, |k|\} \end{aligned}$$

Therefore following the chain of inequalities we get that

$$0 \leq \lim_{(h,k) \rightarrow 0} \frac{|f(h,k)|}{|(h,k)|} \leq \lim_{(h,k) \rightarrow 0} Mnm \min\{|h|, |k|\} = 0$$

Hence

$$\lim_{(h,k) \rightarrow 0} \frac{|f(h,k)|}{|(h,k)|} = 0$$

(b) We want to prove that $Df(a,b)(x,y) = f(a,y) + f(x,b)$, then see that

$$\begin{aligned} &\frac{|f(a+h, b+k) - f(a,b) - Df(a,b)(h,k)|}{|(h,k)|} = \\ &= \frac{|f(a,b) + f(a,k) + f(h,b) + f(h,k) - f(a,b) - Df(a,b)(h,k)|}{|(h,k)|} \\ &= \frac{|f(a,k) + f(h,b) + f(h,k) - Df(a,b)(h,k)|}{|(h,k)|} \\ &= \frac{|f(a,k) + f(h,b) + f(h,k) - (f(a,k) + f(h,b))|}{|(h,k)|} \\ &= \frac{|f(h,k)|}{|(h,k)|} \end{aligned}$$

But we saw on part (a) that

$$\lim_{(h,k) \rightarrow 0} \frac{|f(h,k)|}{|(h,k)|} = 0$$

Therefore $Df(a,b)(x,y) = f(a,y) + f(x,b)$ is the derivative.

(c) From Theorem 2-3 if $p(x,y) = xy$ then we know that

$$Dp(a,b)(x,y) = bx + ay$$

But from part (b) we know that

$$Df(a,b)(x,y) = f(x,b) + f(a,y)$$

So taking $f = p$ we get that

$$Dp(a,b)(x,y) = p(x,b) + p(a,y) = xb + ay$$

Therefore the equation we have for $Dp(a,b)(x,y)$ from Theorem 2-3 is a special case of the equation $Df(a,b)(x,y) = f(x,b) + f(a,y)$.

□

Proof. 2-13

Let $IP : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ be defined as $IP(x, y) = \langle x, y \rangle$

(a) We see that if $x, x_1, x_2, y, y_1, y_2 \in \mathbb{R}^n$ and $a \in \mathbb{R}$ we have that

$$\begin{aligned}\langle ax, y \rangle &= a\langle x, y \rangle = \langle x, ay \rangle \\ \langle x_1 + x_2, y \rangle &= \langle x_1, y \rangle + \langle x_2, y \rangle \\ \langle x, y_1 + y_2 \rangle &= \langle x, y_1 \rangle + \langle x, y_2 \rangle\end{aligned}$$

Then IP is a bilinear function so from Problem 2-12 we have that

$$D(IP(x, y))(a, b) = IP(a, y) + IP(x, b) = \langle a, y \rangle + \langle x, b \rangle$$

To compute the Jacobian matrix $(IP(x, y))'(a, b)$ we assume that $IP(x, y)$ is defined as $IP(x, y) = x_1y_1 + x_2y_2 + \dots + x_ny_n$ where x_i and y_i are the components of x and y . Then

$$(IP(x, y))' = (y_1, y_2, \dots, y_n, x_1, x_2, \dots, x_n)$$

Hence

$$\begin{aligned}(IP(x, y))'(a, b) &= (y_1, y_2, \dots, y_n, x_1, x_2, \dots, x_n) \begin{pmatrix} a_1 \\ \vdots \\ a_n \\ b_1 \\ \vdots \\ b_n \end{pmatrix} \\ &= a_1y_1 + \dots + a_ny_n + b_1x_1 + \dots + b_nx_n \\ &= IP(a, y) + IP(x, b)\end{aligned}$$

(b) Let $f, g : \mathbb{R} \rightarrow \mathbb{R}^n$ be differentiable functions and let $h : \mathbb{R} \rightarrow \mathbb{R}$ be defined by $h(t) = \langle f(t), g(t) \rangle$.

We can write h as $h(t) = IP(f(t), g(t))$, so using the chain rule we get that

$$\begin{aligned}D(h(a))(v) &= D(IP(f(a), g(a)))(Df(a)(v), Dg(a)(v)) \\ &= IP(Df(a)(v), g(a)) + IP(f(a), Dg(a)(v))\end{aligned}$$

Since f, g are differentiable then $Df(a)(v) = f'(a)(v)$ and $Dg(a)(v) = g'(a)(v)$. Taking $v = 1$ we get that

$$h'(a) = \langle f'(a)^T, g(a) \rangle + \langle f(a), g'(a)^T \rangle$$

Where we put $f'(a)^T$ and $g'(a)^T$ because we are considering $IP : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ to be defined as $\langle x, y \rangle = xy$.

- (c) Let $f : \mathbb{R} \rightarrow \mathbb{R}^n$ be differentiable and let $|f(t)| = 1$ for all t .
 Let us consider the function $h(t) = \langle f(t), f(t) \rangle$, then we know that

$$h'(a) = \langle f'(a)^T, f(a) \rangle + \langle f(a), f'(a)^T \rangle$$

But also we know that

$$h(t) = \langle f(t), f(t) \rangle = |f(t)|^2 = 1$$

So we get that $h'(a) = 0$ for any a , and hence

$$\begin{aligned} 0 &= \langle f'(a)^T, f(a) \rangle + \langle f(a), f'(a)^T \rangle \\ \langle f(a), f'(a)^T \rangle &= -\langle f'(a)^T, f(a) \rangle \\ \langle f'(a)^T, f(a) \rangle &= -\langle f'(a)^T, f(a) \rangle \end{aligned}$$

Where we used that $\langle y, x \rangle = \langle x, y \rangle$ if $x, y \in \mathbb{R}^n$. But this can only be true if

$$\langle f'(t)^T, f(t) \rangle = 0$$

- (d) Let us consider the function $f(t) = t$ then the function $|f|$ gives us

$$|f|(t) = |f(t)| = |t|$$

Then we see that

$$\lim_{h \rightarrow 0^+} \frac{|f|(0+h) - |f|(0)}{h} = \lim_{h \rightarrow 0^+} \frac{|h| - |0|}{h} = \lim_{h \rightarrow 0^+} \frac{h}{h} = 1$$

But

$$\lim_{h \rightarrow 0^-} \frac{|f|(0+h) - |f|(0)}{h} = \lim_{h \rightarrow 0^-} \frac{|h| - |0|}{h} = \lim_{h \rightarrow 0^-} \frac{-h}{h} = -1$$

Then $|f|$ is not differentiable at 0.

□

Proof. **2-14**

- (a) Let f be multilinear, $i \neq j$ and $h = (h_1, \dots, h_k)$ with $h_l \in E_l$ then we note that

$$0 \leq \frac{|f(a_1, \dots, h_i, \dots, h_j, \dots, a_k)|}{|h|} \leq \frac{|f(a_1, \dots, h_i, \dots, h_j, \dots, a_k)|}{|(h_i, h_j)|}$$

We see that $f(a_1, \dots, h_i, \dots, h_j, \dots, a_k)$ is bilinear with respect to h_i and h_j , but from problem 2-12 we know that

$$\lim_{(h_i, h_j) \rightarrow 0} \frac{|f(a_1, \dots, h_i, \dots, h_j, \dots, a_k)|}{|(h_i, h_j)|} = 0$$

So by the squeeze theorem we get that

$$\lim_{h \rightarrow 0} \frac{|f(a_1, \dots, h_i, \dots, h_j, \dots, a_k)|}{|h|} = 0$$

- (b) Let note that for the case $k = 4$ we have that

$$\begin{aligned} f(a_1 + h_1, a_2 + h_2, a_3 + h_3, a_4 + h_4) &= \\ &= f(a_1, a_2 + h_2, a_3 + h_3, a_4 + h_4) + f(h_1, a_2 + h_2, a_3 + h_3, a_4 + h_4) \\ &= f(a_1, a_2, a_3 + h_3, a_4 + h_4) + f(a_1, h_2, a_3 + h_3, a_4 + h_4) \\ &\quad + f(h_1, a_2, a_3 + h_3, a_4 + h_4) + f(h_1, h_2, a_3 + h_3, a_4 + h_4) \\ &= f(a_1, a_2, a_3, a_4 + h_4) + f(a_1, a_2, h_3, a_4 + h_4) \\ &\quad + f(a_1, h_2, a_3, a_4 + h_4) + f(a_1, h_2, h_3, a_4 + h_4) \\ &\quad + f(h_1, a_2, a_3, a_4 + h_4) + f(h_1, a_2, h_3, a_4 + h_4) \\ &\quad + f(h_1, h_2, a_3, a_4 + h_4) + f(h_1, h_2, h_3, a_4 + h_4) \\ &= f(a_1, a_2, a_3, a_4) + f(a_1, a_2, a_3, h_4) + f(a_1, a_2, h_3, a_4) + f(a_1, a_2, h_3, h_4) \\ &\quad + f(a_1, h_2, a_3, a_4) + f(a_1, h_2, a_3, h_4) + f(a_1, h_2, h_3, a_4) + f(a_1, h_2, h_3, h_4) \\ &\quad + f(h_1, a_2, a_3, a_4) + f(h_1, a_2, a_3, h_4) + f(h_1, a_2, h_3, a_4) + f(h_1, a_2, h_3, h_4) \\ &\quad + f(h_1, h_2, a_3, a_4) + f(h_1, h_2, a_3, h_4) + f(h_1, h_2, h_3, a_4) + f(h_1, h_2, h_3, h_4) \end{aligned}$$

This is also true for when we take k arguments, where we are summing over all possible choices of a_i and h_i . So, when computing

$$\frac{|f(a_1 + h_1, \dots, a_k + h_k) - f(a_1, \dots, a_k) - Df(a_1, \dots, a_k)h|}{|h|}$$

We note that except for the terms $f(a_1, \dots, h_i, \dots, a_k)$ the rest are bilinear functions of some h_i and h_j and from part (a) we know that

$$\lim_{h \rightarrow 0} \frac{|f(a_1, \dots, h_i, \dots, h_j, \dots, a_k)|}{|h|} = 0$$

So we are left with

$$\begin{aligned} & \frac{|f(a_1 + h_1, \dots, a_k + h_k) - f(a_1, \dots, a_k) - Df(a_1, \dots, a_k)h|}{|h|} = \\ &= \frac{|\sum_{i=1}^k f(a_1, \dots, h_i, \dots, a_k) - Df(a_1, \dots, a_k)h|}{|h|} \end{aligned}$$

Therefore must be that

$$Df(a_1, \dots, a_k)(h_1, \dots, h_k) = \sum_{i=1}^k f(a_1, \dots, h_i, \dots, a_k)$$

Which is possible since each $f(a_1, \dots, h_i, \dots, a_k)$ is a linear function of h_i as it should be.

□

Proof. 2-15

Let us consider an $n \times n$ matrix as a point in the n -fold product $\mathbb{R}^n \times \dots \times \mathbb{R}^n$ by considering each row as a member of $\mathbb{R}^n$.

- (a) Let $\det : \mathbb{R}^n \times \dots \times \mathbb{R}^n \rightarrow \mathbb{R}$ we want to prove it's differentiable.
Let us consider first the case when $n = 3$ then fixing all rows except for the second one we get that

$$\begin{aligned} \det \begin{pmatrix} a_1 \\ x \\ a_3 \end{pmatrix} &= \det \begin{pmatrix} a_{11} & a_{12} & a_{13} \\ x_1 & x_2 & x_3 \\ a_{31} & a_{32} & a_{33} \end{pmatrix} \\ &= a_{11}x_2a_{33} + a_{12}x_3a_{31} + x_1a_{32}a_{13} \\ &\quad - a_{31}x_2a_{13} - a_{32}x_3a_{11} - x_1a_{12}a_{33} \end{aligned}$$

So we see that the determinant is linear with respect to the second row; the same can be shown for all rows.

In the n case, this is still valid, so we see that $\det$ is a multilinear function with respect to its rows, so from problem 2-14 if we let $f = \det$ we get that

$$Df(a_1, \dots, a_n)(x_1, \dots, x_n) = \sum_{i=1}^n f(a_1, \dots, a_{i-1}, x_i, a_{i+1}, \dots, a_n)$$

Hence

$$D(\det)(a_1, \dots, a_n)(x_1, \dots, x_n) = \sum_{i=1}^n \det \begin{pmatrix} a_1 \\ \vdots \\ a_{i-1} \\ x_i \\ a_{i+1} \\ \vdots \\ a_n \end{pmatrix}$$

- (b) Let $a_{ij} : \mathbb{R} \rightarrow \mathbb{R}$ be differentiable and let $f(t) = \det(a_{ij}(t))$.
We note that

$$Df(a_{ij}(t))(v) = Df(a_{ij}(t)) (Da_{ij}(t)(v))$$

Since $a_{ij}(t)$ is differentiable and taking $v = 1$ we get that

$$f'(t) = Df(a_{ij}(t))(1) = Df(a_{ij}(t)) (a'_{ij}(t)) = \sum_{i=1}^n \det \begin{pmatrix} a_{11}(t) & \dots & a_{1n}(t) \\ \vdots & & \vdots \\ a'_{i1}(t) & \dots & a'_{in}(t) \\ \vdots & & \vdots \\ a_{n1}(t) & \dots & a_{nn}(t) \end{pmatrix}$$

(c) We know that $s_j(t)$ is a solution to the equation

$$\sum_{j=1}^n a_{ji}(t)s_j(t) = b_i(t)$$

Since all factors are differentiable we can differentiate both sides as

$$\begin{aligned} \left(\sum_{j=1}^n a_{ji}(t)s_j(t) \right)' &= b_i'(t) \\ a'_{1i}(t)s_1(t) + a_{1i}(t)s'_1(t) + \dots + a'_{ni}(t)s_n(t) + a_{ni}(t)s'_n(t) &= b'_i(t) \\ \sum_{j=1}^n a'_{ji}(t)s_j(t) + \sum_{j=1}^n a_{ji}(t)s'_j(t) &= b'_i(t) \end{aligned}$$

Hence

$$\sum_{j=1}^n a_{ji}(t)s'_j(t) = b'_i(t) - \sum_{j=1}^n a'_{ji}(t)s_j(t)$$

We may write this equation for every i th column, so if we write it in matrix notation as $A(t) = [a_{ji}(t)]$, $b(t) = [b_i(t)]$ and $s(t) = [s_j(t)]$ we get that

$$\begin{aligned} A(t)s'(t) &= b'(t) - A'(t)s(t) \\ s'(t) &= A^{-1}(t)(b'(t) - A'(t)s(t)) \end{aligned}$$

□

Proof. **2-16**

Let $f : \mathbb{R}^n \rightarrow \mathbb{R}^n$ and $f^{-1} : \mathbb{R}^n \rightarrow \mathbb{R}^n$ be differentiable, we want to show that $(f^{-1})'(a) = [f'(f^{-1}(a))]^{-1}$.

Let us notice that $f(f^{-1}(x)) = x$ and since $x \in \mathbb{R}^n$ is also differentiable we can differentiate both sides as follows

$$\begin{aligned}(f(f^{-1}(a)))' &= a' \\ f'(f^{-1}(a))(f^{-1}(a))' &= 1 \\ (f^{-1})'(a) &= \frac{1}{f'(f^{-1}(a))}\end{aligned}$$

Where we used the chain rule on the LHS.

□

Partial Derivatives

Proof. 2-21

Let $g_1, g_2 : \mathbb{R}^2 \rightarrow \mathbb{R}$ be continuous and let us define $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = \int_0^x g_1(t, 0) dt + \int_0^y g_2(x, t) dt$$

(a) We want to compute $D_2 f(x, y)$ then using the Leibniz rule we get that

$$\begin{aligned} D_2 f(x, y) &= D_2 \left(\int_0^x g_1(t, 0) dt \right) + D_2 \left(\int_0^y g_2(x, t) dt \right) \\ &= g_1(x, 0) \cdot 0 - g_1(0, 0) \cdot 0 + \int_0^x D_2 g_1(t, 0) dt \\ &\quad + g_2(x, y) D_2(y) - g_2(x, 0) \cdot 0 + \int_0^y D_2 g_2(x, t) dt \\ &= g_2(x, y) \end{aligned}$$

Where we used that $D_2(x) = D_2(0) = 0$ and that

$$\int_0^x D_2 g_1(t, 0) dt = \int_0^y D_2 g_2(x, t) dt = 0$$

Since none of them depend on y .

(b) If we define f as

$$f(x, y) = \int_0^x g_1(t, y) dt + \int_0^y g_2(0, t) dt$$

Then in the same way as in part (a) we see that

$$\begin{aligned} D_1 f(x, y) &= D_1 \left(\int_0^x g_1(t, y) dt \right) + D_1 \left(\int_0^y g_2(0, t) dt \right) \\ &= g_1(x, y) \cdot D_1(x) - g_1(0, y) \cdot D_1(0) + \int_0^x D_1 g_1(t, y) dt \\ &\quad + g_2(0, y) D_1(y) - g_2(0, 0) \cdot D_1(0) + \int_0^y D_1 g_2(0, t) dt \\ &= g_1(x, y) \end{aligned}$$

(c) Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$f(x, y) = \frac{1}{2}x^2 + \frac{1}{2}y^2$$

Then we see that

$$D_1 f(x, y) = x \quad D_2 f(x, y) = y$$

On the other hand, if we let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined as $f(x, y) = xy$ then we see that

$$D_1 f(x, y) = y \quad D_2 f(x, y) = x$$

□

Proof. 2-22

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ such that $D_2f = 0$, also, let us take $(x, y) \in \mathbb{R}^2$ where x is fixed, then we can define $g(y) = f(x, y)$ i.e. g is a function of 1 variable.

By the Mean Value Theorem taking an interval $[a, b] \in \mathbb{R}$ we know there is $c \in [a, b]$ such that

$$g'(c) = \frac{g(b) - g(a)}{b - a}$$

But since $g'(y) = D_2f(x, y) = 0$ for all $y \in \mathbb{R}$ we have that

$$\begin{aligned} \frac{g(b) - g(a)}{b - a} &= 0 \\ g(b) &= g(a) \end{aligned}$$

But then this implies that $f(x, b) = g(b) = g(a) = f(x, a)$. Since the interval is arbitrary then f is independent of the second variable.

So we may assume that f is of the form $h(x) = f(x, y)$, if we also have that

$$D_1f(x, y) = 0 = h'(x)$$

then, must be that $h(x) = c$ for some constant c . Therefore

$$f(x, y) = c$$

□

Proof. **2-23**

Let $A = \{(x, y) \in \mathbb{R}^2 : x < 0 \text{ or } x \geq 0 \text{ and } y \neq 0\}$

(a) Let $f : A \rightarrow \mathbb{R}$ such that $D_1f = D_2f = 0$.

Suppose we take a point $(a, b) \in A$ where $a, b > 0$ we see that

$$f(a, b) = f(-a, b)$$

because $D_1f = 0$.

In the same way

$$f(-a, b) = f(-a, -b)$$

because $D_2f = 0$.

Hence we can get that

$$f(a, b) = f(-a, b) = f(-a, -b) = f(a, -b)$$

If $a < 0$ and $b > 0$ then we see that

$$f(a, b) = f(-a, b) = f(a, -b) = f(-a, -b)$$

We can show this for all the cases and since a, b are arbitrary then f must be constant in A .

(b) Let us define

$$f(x, y) = \begin{cases} 1 & \text{when } x < 0 \\ 1 & \text{when } x \geq 0, y > 0 \\ 2 & \text{when } x \geq 0, y < 0 \end{cases}$$

Then in all cases we have that $D_2f = 0$ but the function f still depends on y since we get different results for it depending on which side of the positive x -axis we are.

□

Proof. **2-24**

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ defined as

$$f(x, y) = \begin{cases} xy \frac{x^2 - y^2}{x^2 + y^2} & (x, y) \neq 0 \\ 0 & (x, y) = 0 \end{cases}$$

(a) We see that for $(x, y) \neq 0$ we get that

$$\begin{aligned} D_1 f(x, y) &= \frac{y(x^4 + 4x^2y^2 - y^4)}{(x^2 + y^2)^2} \\ D_2 f(x, y) &= \frac{x^5 - 4x^3y^2 - xy^4}{(x^2 + y^2)^2} \end{aligned}$$

Then for $x \neq 0$, $D_2 f(x, 0)$ is

$$D_2 f(x, 0) = \frac{x^5 - 0 - 0}{(x^2 + 0)^2} = \frac{x^5}{x^4} = x$$

If $x = 0$ we see that

$$D_2 f(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, 0 + h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 \cdot h \cdot \frac{0 - h^2}{0 + h^2} - 0}{h} = 0 = x$$

On the other hand, $D_1 f(0, y)$ for $y \neq 0$ is

$$D_1 f(0, y) = \frac{y(0 + 0 - y^4)}{(0 + y^2)^2} = \frac{-y^5}{y^4} = -y$$

And when $y = 0$ we get that

$$D_1 f(0, 0) = \lim_{h \rightarrow 0} \frac{f(0 + h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h \cdot 0 \cdot \frac{h^2 - 0}{h^2 + 0} - 0}{h} = 0 = -y$$

Therefore $D_2 f(x, 0) = x$ for all x and $D_1 f(0, y) = -y$ for all y .

(b) We see that

$$\begin{aligned} D_{1,2}f(0,0) &= \lim_{h \rightarrow 0} \frac{D_2f(0+h,0) - D_2f(0,0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{D_2f(h,0) - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{h-0}{h} \\ &= 1 \end{aligned}$$

Where we used that $D_2f(h,0) = h$ for all h . On the other hand, we have that

$$\begin{aligned} D_{2,1}f(0,0) &= \lim_{h \rightarrow 0} \frac{D_1f(0,0+h) - D_1f(0,0)}{h} \\ &= \lim_{h \rightarrow 0} \frac{D_1f(0,h) - 0}{h} \\ &= \lim_{h \rightarrow 0} \frac{-h-0}{h} \\ &= -1 \end{aligned}$$

Where we used that $D_1f(0,h) = -h$ for all h . Therefore

$$D_{1,2}f(0,0) \neq D_{2,1}f(0,0)$$

□

Proof. 2-25

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be defined as

$$f(x) = \begin{cases} e^{-x^{-2}} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

If $x \neq 0$ then we see that the function $-1/x^2$ is in C^∞ on $\mathbb{R} \setminus \{0\}$ and since the function e^t is in C^∞ as well, then we can conclude that the composition $e^{-x^{-2}}$ is in C^∞ .

If $x = 0$ we see that

$$f'(0) = \lim_{h \rightarrow 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \rightarrow 0} \frac{e^{-h^{-2}}}{h} = \lim_{h \rightarrow 0} \frac{1/h}{e^{h^{-2}}}$$

Since we have an indeterminate limit we can use L'Hopital rule as follows

$$\lim_{h \rightarrow 0} \frac{1/h}{e^{h^{-2}}} = \lim_{h \rightarrow 0} \frac{-1/h^2}{-2e^{h^{-2}}/h^3} = \lim_{h \rightarrow 0} \frac{h}{2e^{h^{-2}}} = 0$$

Now let us compute $f''(0)$ as follows

$$f''(0) = \lim_{h \rightarrow 0} \frac{f'(0+h) - f'(0)}{h} = \lim_{h \rightarrow 0} \frac{2e^{-h^{-2}}/h^3}{h} = \lim_{h \rightarrow 0} \frac{2/h^4}{e^{h^{-2}}}$$

Applying L'Hopital rule twice, we get that

$$\begin{aligned} \lim_{h \rightarrow 0} \frac{2/h^4}{e^{h^{-2}}} &= \lim_{h \rightarrow 0} \frac{-8/h^5}{-2e^{h^{-2}}/h^3} \\ &= \lim_{h \rightarrow 0} \frac{4/h^2}{e^{h^{-2}}} \\ &= \lim_{h \rightarrow 0} \frac{-8/h^3}{-2e^{h^{-2}}/h^3} \\ &= \lim_{h \rightarrow 0} \frac{4}{e^{h^{-2}}} \\ &= 0 \end{aligned}$$

We see that every derivative of $e^{-x^{-2}}$ when $x \neq 0$ is of the form

$$f^{(i)}(x) = \frac{P(x)}{Q(x)} e^{-x^{-2}}$$

Where $P(x)$ and $Q(x)$ are polynomials. So when computing the derivatives at $x = 0$ we always get something like

$$f^{(i)}(x) = \lim_{h \rightarrow 0} \frac{f^{(i-1)}(h)}{h} = \frac{P(h)}{Q(h)} \frac{1/h}{e^{h^{-2}}}$$

We may solve this by applying L'Hopital rule a few times but we also notice that the growth of $e^{h^{-2}}$ is always faster than any polynomial, so the limit will always tend to $1/\infty = 0$. $\square$

Proof. **2-26**

Let $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$f(x) = \begin{cases} e^{-(x-1)^{-2}} e^{-(x+1)^{-2}} & x \in (-1, 1) \\ 0 & x \notin (-1, 1) \end{cases}$$

(a) If $x \in (-1, 1)$ we see that the expression can be written as

$$e^{-(x-1)^{-2}} e^{-(x+1)^{-2}} = e^{-\frac{2(x^2+1)}{(x-1)^2(x+1)^2}}$$

Since $x \in (-1, 1)$ we know the function $-\frac{2(x^2+1)}{(x-1)^2(x+1)^2}$ is in C^∞ and since e^t is also in C^∞ the the composition is in C^∞ .

If $x \in (1, \infty)$ or $x \in (-\infty, -1)$ then the function is 0 so f is in C^∞ there.

The only points where f might not be infinitely differentiable are $x = -1$ and $x = 1$. Suppose $x = -1$ then we see that

$$\begin{aligned} f'(-1) &= \lim_{h \rightarrow 0} \frac{f(-1+h) - f(-1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{e^{-(h-2)^{-2}} e^{-h^{-2}}}{h} \\ &= \lim_{h \rightarrow 0} e^{-(h-2)^{-2}} \lim_{h \rightarrow 0} \frac{e^{-h^{-2}}}{h} \\ &= e^{-x^{-4}} \lim_{h \rightarrow 0} \frac{e^{-h^{-2}}}{h} \\ &= 0 \end{aligned}$$

Where we used the result we got from problem 2-25. Also, from problem 2-25 we get that

$$f^{(i)}(-1) = 0$$

For all i .

On the other hand, if $x = 1$ we get that

$$\begin{aligned} f'(1) &= \lim_{h \rightarrow 0} \frac{f(1+h) - f(1)}{h} \\ &= \lim_{h \rightarrow 0} \frac{e^{-h^{-2}} e^{-(h+2)^{-2}}}{h} \\ &= \lim_{h \rightarrow 0} e^{-(h+2)^{-2}} \lim_{h \rightarrow 0} \frac{e^{-h^{-2}}}{h} \\ &= e^{-x^{-4}} \lim_{h \rightarrow 0} \frac{e^{-h^{-2}}}{h} \\ &= 0 \end{aligned}$$

So we see that also for all i we get that $f^{(i)}(1) = 0$.

Therefore the function f is in C^∞ .

If $x = 0$ we see that

$$f(0) = e^{-(-1)^{-2}} e^{-(1)^{-2}} = \frac{1}{e^{\frac{1}{(-1)^2}}} \frac{1}{e^{\frac{1}{1^2}}} = \frac{1}{e^2} > 0$$

If $x \in (-1, 1) \setminus \{0\}$ the function is given by

$$f(x) = \frac{1}{e^{\frac{1}{(x-1)^2}}} \frac{1}{e^{\frac{1}{(x+1)^2}}}$$

But $(x-1)^2$ and $(x+1)^2$ are positive always so the function is always positive here.

Therefore the function f is positive in the interval $(-1, 1)$

(b) Let us consider the function

$$g(x) = \frac{\int_0^x f}{\int_0^\epsilon f}$$

Where f is a function C^∞ which is positive on $(0, \epsilon)$ and 0 elsewhere.

If $x \leq 0$ we see that $\int_0^x f = 0$ and $\int_0^\epsilon f > 0$ then $g(x) = 0$.

Now, suppose $x \geq \epsilon$ then $\int_0^x f = \int_0^\epsilon f$ since f is positive on $(0, \epsilon)$ and 0 elsewhere, hence

$$g(x) = \frac{\int_0^x f}{\int_0^\epsilon f} = \frac{\int_0^\epsilon f}{\int_0^\epsilon f} = 1$$

We need to prove now that g is in C^∞ .

We note that

$$g'(x) = \frac{(\int_0^x f)'}{\int_0^\epsilon f} = \frac{f'}{\int_0^\epsilon f}$$

Where we used the Fundamental Theorem of Calculus. Since $f \in C^\infty$ then the expression for g' is also infinitely differentiate, so $g \in C^\infty$.

(c) Let f be defined as

$$f(x) = \begin{cases} e^{-(x-1)^{-2}} e^{-(x+1)^{-2}} & x \in (-1, 1) \\ 0 & x \notin (-1, 1) \end{cases}$$

Also, we define $g : \mathbb{R}^n \rightarrow \mathbb{R}$ as

$$g(x) = f([x^1 - a^1]/\epsilon) \cdot \dots \cdot f([x^n - a^n]/\epsilon)$$

Where $a \in \mathbb{R}^n$. We know that f is in C^∞ because of part (a) then the multiplication of f n-times is in C^∞ so g is in C^∞ .

Let $x \in (a^1 - \epsilon, a^1 + \epsilon) \times \dots \times (a^n - \epsilon, a^n + \epsilon)$ then we see that

$$-1 \leq \frac{x^i - a^i}{\epsilon} \leq 1$$

for $i \in \{1, \dots, n\}$. We know that in the interval $(-1, 1)$, f is positive then so is g .

We know that elsewhere f is 0 so g is also 0 elsewhere.

(d) Let $A \subset \mathbb{R}^n$ be open and $C \subset A$ be compact. Let us take for every point $c \in C$ an open box

$$(c^1 - \epsilon, c^1 + \epsilon) \times \dots \times (c^n - \epsilon, c^n + \epsilon)$$

where ϵ is sufficiently small so the open boxes do not intersect with the closed set B where f is supposed to be 0.

This is an open cover of C and since C is compact must be a finite subcover that covers C as well.

Then we can define a set of functions $g_i : \cup_{i=1}^m O_i \rightarrow \mathbb{R}$ over the finite subcover $\cup_{i=1}^m O_i$ such that g_i is defined as

$$g_i(x) = \begin{cases} f([x^1 - o_i^1]/\epsilon) \cdot \dots \cdot f([x^n - o_i^n]/\epsilon) & x \in O_i \\ 0 & x \notin O_i \end{cases}$$

Where o_i is the center of the open box O_i and the function f is defined as in part (c).

If we add all the functions we get another function $g : \cup_{i=1}^m O_i \rightarrow \mathbb{R}$ defined as

$$g(x) = \sum_{i=1}^m g_i(x)$$

We note that g is defined in an open set $A = \cup_{i=1}^m O_i$, also g is in C^∞ because each g_i is in C^∞ and $g(x) > 0$ for $x \in C$ but $g = 0$ at B , the closed set.

- (e) Since the function g defined in part (d) has the property $g(x) > 0$ for every $x \in C$ then we can always choose an $\epsilon > 0$ such that $g(x) \geq \epsilon$. So if we take the composed function $f = h \circ g$ where h is defined as in part (b) i.e. $h : \mathbb{R} \rightarrow [0, 1]$ such that $h(x) = 0$ for $x \leq 0$ and $h(x) = 1$ for $x \geq \epsilon$ then we see that $f = 1$ for every $x \in C$.

□